Claire C. Winfrey ~ *Curriculum Vitae*

Address:
University of Colorado Boulder
Department of Ecology and Evolutionary Biology
1900 Pleasant Street
334 UCB
Boulder, CO 80309-0334
Email: claire.winfrey@colorado.edu
Website: clairecwinfrey.github.io

EDUCATION

Present – 2020	**PhD in Ecology & Evolutionary Biology (EBIO)** (GPA: 4.00), *anticipated Fall 2026* University of Colorado Boulder (CU) Co-advisors: Dr. Noah Fierer and Dr. Julian Resasco
2020 – 2017	**Master of Science in Ecology & Evolutionary Biology (EEB),** (GPA: 3.99) University of Tennessee, Knoxville (UTK) Advisor: Dr. Kimberly Sheldon
2016 – 2011	**B.S. in Biology; minors in Spanish, Linguistics,** *Summa cum laude* (GPA: 3.80) University of Oklahoma (OU) Honors Thesis Advisor: Dr. Ola Fincke

PEER-REVIEWED PUBLICATIONS (Undergraduate mentee is underlined)

7. **Winfrey, C.C.**, Socualaya-Torres, A., Resasco, J., & N. Fierer. The sources and traits of bacteria and fungi found in the near-surface atmosphere. Accepted in *Applied and Environmental Microbiology.*

6. **Winfrey, C.C.**, Resasco, J., & N. Fierer. (2025). Habitat specialization and edge effects of soil microbial communities in a fragmented landscape, *Ecology, 106(4)*. doi: 10.1002/ecy.70072

5. Ramoneda, J., Stallard-Olivera, E., Hoffert, M., **Winfrey, C.C.**, Stadler, M., Niño-García, J.P., & Fierer, N. (2023). *Science Advances, 9(17).* Building a genome-based understanding of bacterial pH preferences. doi: 10.1126/sciadv.adf8998

4. Dickey, J.R., Swenie, R.A., Turner, S.C., **Winfrey, C.C.**, Yaffar, D., Padukone, A., Beals, K.K., Sheldon, K.S., & S.N. Kivlin. (2021). The utility of macroecological rules for microbial biogeography. *Frontiers in Ecology and Evolution*, *9.* doi: 10.3389/fevo.2021.633155

3. **Winfrey, C.** & Fincke, O. M. (2017). Role of visual and non-visual cues in damselfly mate recognition. *International Journal of Odonatology*, 20, 43–52. doi: 10.1080/13887890.2017.1297259

2. Marhanka, E.C., J.L Watters, N.A. Huron, S.L. McMillin, **C.C. Winfrey**, D.J. Curtis, D.R. Davis, J.K. Farkas, J.L. Kerby, and C.D. Siler. (2017). Detection of high prevalence of *Batrachochytrium dendrobatidis* in amphibians from southern Oklahoma, USA. *Herpetological Review,* 48(1), 70-74.

1. Davis, D. R., A. D. Geheber, J. L. Watters, M. L. Penrod, K. D. Feller, A. Ashford, J. Kouri, D. Nguyen, K. Shauberger, K. Sheatsley, **C. Winfrey**, R. Wong, M. B. Sanguila, R. M. Brown & C. D. Siler. (2016). Additions to Philippine Slender Skinks of the *Brachymeles bonitae* Complex (Reptilia:Squamata:Scincidae) III: a new species from Tablas Island. *Zootaxa,* 4132, (1), 30-43. http://dx.doi.org/10.11646/zootaxa.4132.1.3

IN-PREPARATION PUBLICATIONS

1. **Winfrey, C.C.,** VanderBurgh, C., Zahid, A.M., Trivedi, P., & N. Fierer. *In preparation*. The diversity and distribution of bacteria and archaea that form spore-like structures in the soil.

2. Hoffert, M., Wilson, J., **Winfrey, C.C.** Coffman, M., Ramoneda, J., Nymann, T., Yeo, E., Clark, K., Quispe, R., Collins, J., Biegert, J., Gavin, M., Dunn, R.R., & N. Fierer. *In preparation*. Experimental study of microbial dynamics in wild-fermented beers.

FELLOWSHIPS, GRANTS, & AWARDS

Present – 2025	**Cooperative Institute for Research in Environmental Sciences (CIRES) Graduate Student Research Award** (funds a 50% research appointment in CIRES for one year, $29,970)
2026	**Boulder County Nature Association Grants for Research** ($5,000)
2026	CIRES Graduate Student Travel Award ($750)
2025	CU EBIO Graduate Student Research Fund ($2,000)
2024 – 2019	**National Science Foundation Graduate Research Fellowship** ($138,000)
2024	CU Graduate School Travel Grant ($700)
2023	CU Graduate and Professional Student Government Travel Grant ($300)
2022	CIRES Graduate Student Travel Award ($1,500)
2021	CU EBIO Graduate Student Research Fund ($2,200)
2021 – 2020	CU Boulder Graduate School Diversity Recruitment Fellowship ($10,000)
2020 – 2017	Tennessee Fellowship for Graduate Excellence ($40,000)
2020	UTK Graduate Student Senate Travel Award ($830)
2019 – 2017	**Program for Excellence & Equity in Research (PEER) Fellowship** (internal UTK fellowship funded by NIH IMSD; tuition and $24,000/year stipend for 2 years)
2019	UTK Graduate Student Senate Travel Award ($500)
2019	Sigma Xi Grants in Aid of Research ($500)
2019	The Coleopterists Society Graduate Student Research Enhancement Award ($1,989)
2018	UTK EEB Departmental Research Funds ($1,396)
2016	Friends of the University of Oklahoma Biological Station Scholarship ($400)
2015	Harley P. Brown Scholarship ($400)

PRESENTATIONS (**invited)*

2026	**Winfrey, C.C.**, VanderBurgh, C., Zahid, A.M., Trivedi, P., & Fierer, N. *The diversity and distribution of soil bacteria capable of forming spore−like structures.* Oral presentation, Soil Ecology Society Biennial Meeting, Southampton Township, NJ, United States.
2025	*** Winfrey, C.C.** *Microbes on the landscape scale: Insights into spatial patterns and traits of soil and airborne fungi and bacteria*. Oral presentation given as a seminar at Kellogg Biological Station, Hickory Corners, MI, United States.
2024	**Winfrey, C.C.**, Resasco, J., Torres, A., & Fierer, N. *Patterns and sources of the air microbiome in an experimentally-fragmented landscape.* Poster presentation, ISME19 (19th International Symposium on Microbial Ecology), Cape Town, South Africa.
2023	**Winfrey, C.C.**, Resasco, J., Torres, A., & Fierer, N. *Patterns and sources of the air microbiome in an experimentally-fragmented landscape.* Poster presentation, Front Range Microbiome Symposium, Fort Collins, CO, United States.
2022	**Winfrey, C.C.,** Resasco, J., & Fierer, N. *Habitat fragmentation and soil biodiversity: Do soil microbes exhibit edge effects?* Oral presentation, Ecological Society of America Annual Meeting, Montréal, QC, Canada.
2020	*** Winfrey, C.C.** *How range overlap and environmental variation influence the gut microbiomes of Phanaeus vindex and P. difformis dung beetles*. Oral presentation, Coleopterist Society Annual Meeting, virtual (web-based) conference.
2020	***Winfrey, C.C.** & Sheldon, K.S. *Understanding spatial variation in the vertically-transmitted gut microbiome of sympatric species of dung beetles*. Oral presentation, Entomological Society of America Annual Meeting, virtual (web-based) conference.

2020 **Winfrey, C.C.** & Sheldon, K.S. *Understanding spatial variation in the vertically-transmitted gut microbiome of sympatric species of dung beetles*. Oral presentation, Ecological Society of America Annual Meeting, virtual (web-based) conference.

2019 **Winfrey, C.C.**, & Sheldon, K.S. *Sister species of dung beetles show species-specific gut microbiota in sympatry*. Poster presentation, Ecological Society of America Annual Meeting, Louisville, KY, United States.

2016 **Winfrey, C.,** Fincke, O.M. *You'll know her when you see her: the role of visual and non-visual cues in damselfly mate recognition*. Oral presentation, Central Ecology and Evolution Conference, Norman, Oklahoma, United States.

2016 **Winfrey, C.**, Fincke, O.M. *You'll know her when you see her: the role of visual and non-visual cues in damselfly mate recognition*. Oral presentation, University of Oklahoma Undergraduate Research Day, Norman, Oklahoma, United States.

MENTORSHIP & TEACHING

2025 **Graduate Core Ecology (University of Colorado), guest teaching assistant.** Aided primary instructor Dr. Katie Suding and two visiting lecturers in developing two units of the graduate-level core ecology class.

2024 Microbiome Amplicon Data Analysis Workshop (hosted by Jessica Metcalf Lab at Colorado State University), workshop teaching assistant.

2024, 2023 **Principles of Ecology (University of Colorado), teaching assistant**. Primary instructor for semester-long lab section of course.

2022 **NSF Research Experiences for Undergraduates (REU)**, primary research mentor for Andrea Torres.

2021 **Research Experiences for Community College Students** (RECCS, funded as NSF REU), primary research mentor for Daniel De Souza.

2020 – 2017 UTK Ecology & Evolutionary Biology Undergraduate Mentorship Program, mentor.

2016 Field Herpetology (University of Oklahoma), teaching assistant for summer session.

OUTREACH & VOLUNTEERING

2026 **CU Boulder EBIO Undergraduate Research Day**, Boulder, CO. Volunteer judge of undergraduate research talks.

2026 "Girls & Science" at Second Saturday at Spur, Colorado State University Spur Campus, Denver, CO. Volunteered to teach children about aerobiology and microscopy.

2025 National Western Stock Show, Denver, CO. Volunteered to teach children and their families about aerobiology and microscopy.

2024 Pollination Celebration at the Gardens on Spring Creek, Fort Collins, CO. Volunteered to teach children and their families about aerobiology and microscopy.

2023 YES!Fest, Greeley, Colorado LINC Library. Volunteered to teach children and their families about aerobiology and microscopy.

2021 Front Range Watershed Days BioBlitz, Estes Park, CO. Volunteered with Left Hand Watershed Center as benthic macroinvertebrate survey leader.

2019 – 2018 UTK Darwin Day, co-President. Organized events to celebrate Darwin and evolutionary science including public talks and various evolution-themed activities and events.

2019 **UTK KidsU: Riveting Reptiles & Awesome Amphibians.** Volunteer co-instructor and course designer of two-week course for middle school students.

2019 Robbinsville Middle School Science Day (at UTK). Designed & led activity on invasive species.

SCIENCE COMMUNICATION

2025 Winfrey, C. "Habitat fragmentation and edges matter for soil microorganisms"*.* *Conservation Corridor*. Science communication article for general audiences and natural resource managers. Available online.

2022 Winfrey, C. "What does landscape connectivity look like for microorganisms?" *Conservation Corridor*. Science communication article for general audiences and natural resource managers. Available online.

2020 Winfrey, C. *How the humble dung beetle can help save the world.* Oral presentation for general audiences. Knoxville Science Forum, University of Tennessee, Knoxville, TN.

2019 Winfrey, C. *How the humble dung beetle can help save the world*. Oral presentation for general audiences. Nova Science Café, Knoxville, TN.

2018 Winfrey, C. *Answers below our feet: soil, health and the effects of soil degradation.* Oral presentation for general audiences. Knoxville Earth Day Fest, Knoxville, TN.

2018. Winfrey, C. *400,000 species strong: how beetles shape ecosystems.* Oral presentation for general audiences. Discover Life in America: Science at Sugarlands. Great Smoky Mountains National Park, Sevier County, TN.

PROFESSIONAL SERVICE

Journal Peer Reviewer: *Ecology Letters, Journal of Animal Ecology*, *Molecular Ecology, PLoS ONE*

Professional Associations: Ecological Society of America, Soil Ecology Society, American Society for Microbiology

Present – 2025 **Soil Ecology Society Student Representative**
2024 – 2022 CU EBIO department colloquium committee, member
2022 – 2021 **Strategies for Ecology Education, Diversity, and Sustainability (SEEDS),** CU Boulder chapter, graduate student advisor
2021 – 2020 CU EBIO outreach committee for Diversity, Equity, & Inclusion, member
2019 Engaging Knoxville in Ecology & Evolution (EKEE), Special Events Coordinator
2019 – 2018 UTK Ecology & Evolutionary Biology Seminar Committee, member
2016 – 2015 OU Biology Student Advisory Committee, faculty-recommended member

HONORS

2019 University of Tennessee EEB Outstanding Outreach and Community Service
2017 The Phi Beta Kappa Society (Alpha of Oklahoma Chapter)
2016 University of Oklahoma Outstanding Biology Undergraduate Award

RELEVANT SKILLS

Bioinformatics and Statistics: environmental amplicon sequencing analysis, multivariate data analysis, linear modeling (general, generalized, mixed, and multilevel), read-based metagenomics, differential abundance analysis, visualization of ecological data

Computational: R (expert), Git version control (proficient), Bash/Unix (familiar), qiime2 platform, high-performance computing on servers and clusters (e.g., *Alpine* supercomputer)

Fieldwork: field collection of soil, bioaerosols, plants, and insects for microbial community analysis, basic vegetation measurements, basic insect and plant identification, coordination of fieldwork logistics

Laboratory Techniques: DNA extraction, PCR, qPCR, library preparation for next-generation sequencing, optimization of yield from diverse environmental samples (soil, phyllosphere, bioaerosol, insect GI tracts, low biomass samples), soil physicochemical analysis (pH, moisture, water holding capacity)